

Random Lipschitz Functions on Graphs with Bounded Expansion

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(joint with Jinyoung Park)

Stanford University

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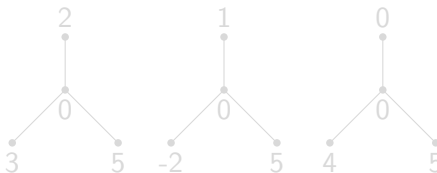
Outline

- 1 Introduction
 - Our Goal
 - Motivation
- 2 Low Expansion
 - Results
 - Proof Idea
- 3 High Local Expansion
 - Result & Next Steps
 - Proof idea

Definitions

- Take any graph G . Let v_0 be a fixed vertex in a graph G and let M be a positive integer.
- Say $\text{Lip}_{v_0}(G; M)$ is the family of maps $f : V(G) \rightarrow \mathbb{Z}$ such that $f(v_0) = 0$ and $|f(u) - f(w)| \leq M$ if $u \sim w$ in G .

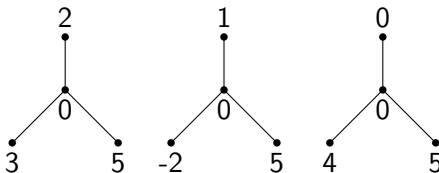
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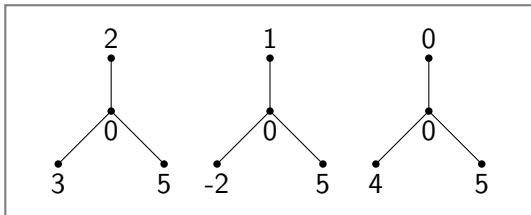
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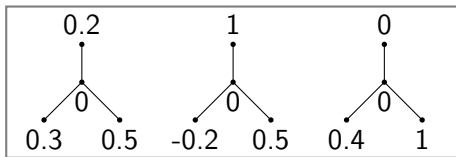
- Sample f from $\text{Lip}_{v_0}(G; M)$ *uniformly at random*:

$$f \in_R \text{Lip}_{v_0}(G; M).$$



Our Goal

- ▶ We want to understand **real-valued Lipschitz functions!**
- ▶ Define $\text{Lip}_{v_0}(G; \infty)$ as the family of maps $f : V(G) \rightarrow \mathbb{R}$ such that $f(v_0) = 0$ and $|f(u) - f(w)| \leq 1$ if $u \sim w$ in G .

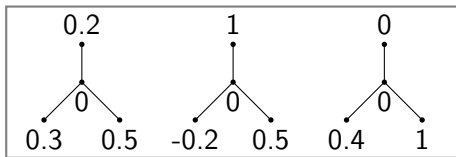


- ▶ Roadmap: $\lim_{M \rightarrow \infty} \left\{ \frac{f}{M} \mid f \in \text{Lip}_{v_0}(G; M) \right\} = \text{Lip}_{v_0}(G; \infty)$

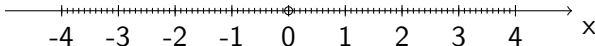


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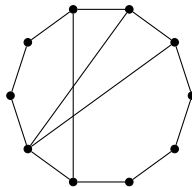
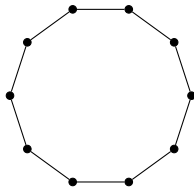
Research Question

Lipschitz functions do not fluctuate much locally.
What about globally?

Primary Interest

The **range** of a (Lipschitz) function f on a connected graph G is

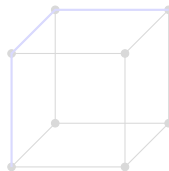
$$R(f) = \sup_{u,v \in G} \{f(u) - f(v)\}$$



Research Question

Can we give a lower or upper bound for a *typical* $R(f)$?

- ▶ Diameter¹ of a graph gives a natural upper bound.



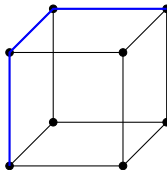
- ▶ We can give tighter bounds that hold with high probability for a uniformly chosen random M -Lipschitz function!

¹The maximum distance between any pair of vertices in a graph.

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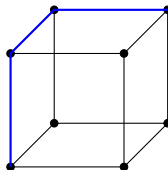
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Background

- ▶ For a d -dimensional hypercube with n nodes, the diameter is $d = \log_2(n)$ but Kahn and Galvin showed that²³

$$\mathbb{P}[R(f) \leq 5] \rightarrow 1 \text{ as } d \rightarrow \infty$$

where f is a random $\mathbb{Z}$ -homomorphism.

- ▶ A $\mathbb{Z}$ -homomorphism f on G such that $f(v_0) = 0$ and $|f(u) - f(w)| = 1$ if $u \sim w$ in G .

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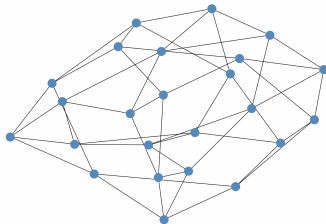
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Background

- Expanders⁴ are locally tree-like but globally have many connections!



⁴a sparse graph that has strong connectivity properties

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- ▶ For a d -regular λ -good expander,⁵ the diameter is $O(\log(n))$.
- ▶ but Peled, Samotij, Yehudayoff showed that⁶

$$\mathbb{P}[R(f) \leq C_{d,\lambda} M \log(\log(n))] \rightarrow 1 \text{ as } n \rightarrow \infty$$

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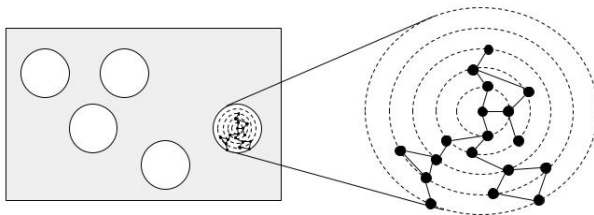
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Expansion Criteria

$$\max_{v \in G} |B_r(v)| \leq c \log(n)$$



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- Previously, expanders were studied: If G is a d -regular λ -expander, then, for *all* r ,

$$|B_{\log(\log(n))}(v)| \geq \log(n)^{2\log(\frac{d}{2\lambda})}$$

- Our assumption has the opposite flavor. We assume **low expansion**: There is a constant $c > 0$,

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Main Result

Theorem (I., Park)

There is a constant $c > 0$ for which the following holds. Let G be a connected graph on n vertices. Fix $v_0 \in V(G)$, and let $f \in_R \text{Lip}_{v_0}(G; M)$. Then for any r that satisfies $\max_{v \in G} |B_r(v)| \leq c \log(n)$,

$$\Pr \left[R(f) < \frac{Mr}{2} \right] = o_n(1).$$

► As long as $d \ll \log(n)$, our bound is not trivial.

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Corollary

Corollary (I., Park)

Let G be a connected λ -expander where $\lambda \leq \frac{d}{32(M+1)\log(9Md^2)}$, and $v_0 \in V(G)$. If $f \in_R \text{Lip}_{v_0}(G; M)$, then

$$R(f) = O\left(M \left\lceil \frac{\log \log n}{\log(d/\lambda)} \right\rceil\right)$$

w.h.p. as $n \rightarrow \infty$.

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Methodology

- ▶ Benjamini, Yadin, and Yehudayoff⁷ gave a similar bound for $\mathbb{Z}$ -homomorphism on graphs.
- ▶ We generalize their results to M -Lipschitz functions. On a high level, the proof follows a similar pattern!

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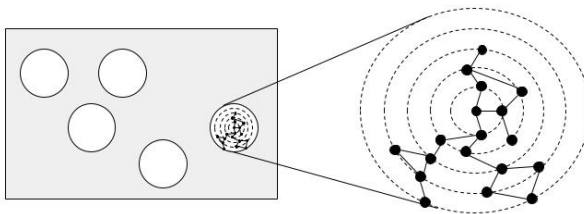
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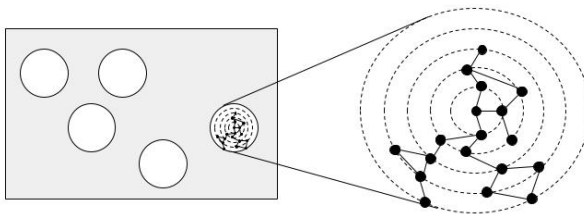
- We know $\sup_{v \in G} |B_r(v)|$ is small. Using this, conclude that there are many disjoint balls of radius r (proof is by induction).



- Then, lower bound the probability a typical Lipschitz function grows quickly on a ball.

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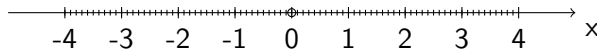
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Remark

Roadmap: $\lim_{M \rightarrow \infty} \left\{ \frac{f}{M} \mid f \in \text{Lip}_{v_0}(G; M) \right\} = \text{Lip}_{v_0}(G; \infty)$



A naive application of their methods renders,

$$\Pr[R(f) \geq C_M r] \rightarrow 1$$

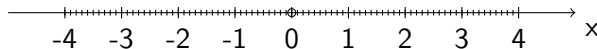
where, importantly, C depends on M . This conclusion does not generalize to real-valued Lipschitz functions because

$$C_M \rightarrow 0 \text{ as } M \rightarrow \infty$$

Our constants do not depend on M !

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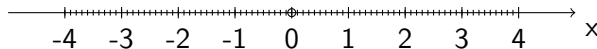
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Methodology continued

- ▶ P is the set of Lipschitz functions on the ball with a high range.
- ▶ Q is the set of all Lipschitz functions.

We want to lower bound P/Q . Now, it is a counting problem!

To do so, we find a set C and mappings T from C to P and Q such that

- ▶ C partitions P : for all $c \neq c' \in C$

$$T_P(c) \cap T_P(c') = \emptyset$$



$$\frac{P}{Q} \geq \inf_{c \in C} \frac{T_P(c)}{T_Q(c)}$$

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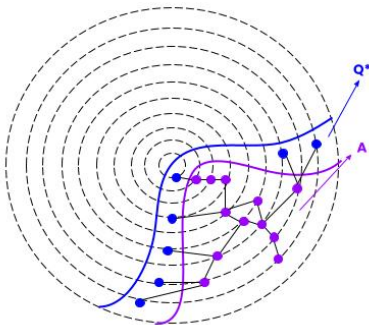
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Mapping

Our plan is to construct many M -Lipschitz functions g on the ball with the following construction:



Mapping

Take some $f \in_R \text{Lip}_{v_0}(G; M)$.

$$g(u) \begin{cases} = f(u) & \text{if } u \in A \cup Q^* \cup \Gamma \\ \in [\frac{Mi}{2}, \frac{M(i+1)}{2} - 1] & \text{if } u \in L_i \setminus (A \cup Q^* \cup \Gamma) \end{cases}$$

Expansion Criteria revisited

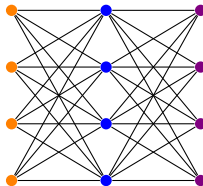
Is this necessary?

$$\max_{v \in G} |B_r(v)| \leq c \log(n)$$

What happens when $d \gg \log(n)$?

Local vs. Global Expansion

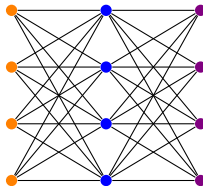
- ▶ We explore $C_{n,k}$ for this. $C_{n,k}$ is a cycle of n layers, where each layer has k vertices and is connected to both its adjacent layers by a complete bipartite graph.



- ▶ It is very well-connected *locally* due to complete bipartites – **large expansion** – but this is not the case *globally*!
- ▶ What is going to happen?

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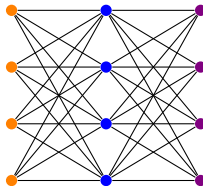
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Background

- If $k \gg \log(n)$, then

$$\Pr[R(f) \leq 3] \rightarrow 1$$

as $n \rightarrow \infty$ where f is a $\mathbb{Z}$ -homomorphism (a type of 1-Lipschitz function) chosen uniformly at random.⁸

- Can we generalize this to M -Lipschitz case?

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Result

Theorem (I., Park)

There is a constant C for which the following holds. For any $\gamma > C$, if $k > \gamma M^2 \log(Mn)$, then for $f \in_R \text{Lip}_{v_0}(C_{n,k}; M)$,

$$\Pr(R(f) \leq M) = 1 - O(1/\gamma).$$

Furthermore, if $\log M = o_n(\log n)$, then

$$\Pr(R(f) \leq M) = 1 - o_n(1).$$

Next Steps

- ▶ What happens in other graphs? Hypercube when $d = \theta(\log_2(n))$?
- ▶ How to generalize our method to real-valued Lipschitz functions? The regime $M \gg n$ is tricky...

Entropy

The entropy of a random variable X , $H(X)$, is a measure of its degree of randomness!

- ▶ The entropy $H(X)$ of X is given by

$$H(X) = \sum_x -p(x) \log p(x),$$

- ▶ For random variable X ,

$$H(X) \leq \log|\Omega|$$

- ▶ (Maximality of the uniform) If X is uniform,

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Background

- ▶ Entropy has been used for counting graph homomorphism from the hypercube⁹ and the general tori¹⁰ before. (One can think of M -Lipschitz functions on a graph G as a homomorphism from G to an infinite path where two vertices are connected if the distance between them is at most M)



- ▶ We adapt their methods for $C_{n,k}$. Challenges:
 - ▶ Galvin et. al. assume the target graph is bounded whereas in our case, it is infinite.
 - ▶ Kahn works with $\mathbb{Z}$ -homomorphisms.

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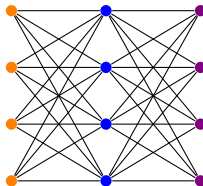


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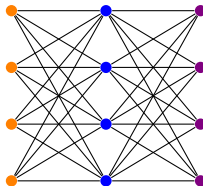
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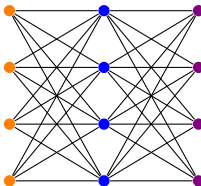
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- ▶ Non-constant layers (when the range is larger than M) restrict the adjacent layers due to the Lipschitz condition.
- ▶ Entropy is maximized when the range of functions on each layer is M .
- ▶ If the range of the function in each layer is M , the typical M -Lipschitz function is flat globally.

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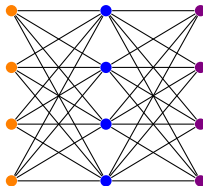
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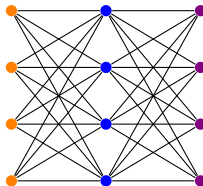
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- ▶ Non-constant layers (when the range is larger than M) restrict the adjacent layers due to the Lipschitz condition.
- ▶ Entropy is maximized when the range of functions on each layer is M .
- ▶ If the range of the function in each layer is M , the typical M -Lipschitz function is flat globally.

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Key Ingredient

Lemma (Shearer's Lemma)

Let $X = (X_1, \dots, X_N)$ be a random vector, and suppose $\alpha : 2^{[N]} \rightarrow \mathbb{R}^+$ satisfies

$$\sum_{A \ni i} \alpha_A = 1 \quad \forall i \in [N].$$

Then for any partial order $\prec$ on $[N]$, we have

$$H(X) \leq \sum_{A \subseteq [N]} \alpha_A H(X_A | (X_i : i \prec A)),$$

where $X_A = (X_i : i \in A)$ and $i \prec A$ means $i \prec a$ for all $a \in A$.

Bounding Entropy

Let L_0 be the layer that v_0 belongs to. For $i \geq 1$, let L_i , the i -th layer, be the set of vertices u such that $(u, v_0) = i$. Define the order $\prec$ on the set of layers as the following:

$$L_1 \prec L_0 \prec L_3 \prec L_2 \prec L_5 \prec L_4 \prec \dots$$

Now using Shearer's Lemma,

$$\begin{aligned} H(f) &\leq \sum_{v \text{ even}, v \neq v_0} H(f(v) | f(x) : x \prec v) \\ &\quad + \frac{1}{2k} \sum_{v \text{ even}} H(f(N_v) | f(x) : x \prec N_v). \end{aligned}$$

We also have $H(f) \geq (nk - 1) \log(M + 1)$. Using both bounds, we derive the result.

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Thank you for listening! Any questions?